# Gauge Alignment and Block-Hopping Locality in a Controlled Rank-Two Periodic One-Dimensional Composite Band

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Draft technical report — November 2, 2026

**Abstract**

A composite band is represented by a matrix-valued Hamiltonian whose real-space block hoppings depend on the reciprocal-space frame chosen inside the retained subspace. We study that dependence in an exact four-state, one-dimensional periodic parent with a rank-two retained spectral space. A polar-transported reference frame is subjected to a prescribed smooth right-unitary rotation, then compared with pointwise and one-global-unitary Procrustes alignment. Complete 64-point block Fourier transforms and symmetric range truncations are assessed on both the construction mesh and an exactly disjoint 257-point staggered mesh. The sampled retained/excluded gap and transport-overlap gates pass. Pointwise alignment recovers reference frames and represented operators at binary64 roundoff, whereas one global unitary does not. The reference and pointwise channels have rapidly decaying block hoppings and reach roundoff-scale spectral truncation errors by range eight. The attacked channel retains an evaluation spectral error of approximately $7.04\times10^{-5}E_{\mathrm{ref}}$ at the same range despite having the same exact pointwise spectra before truncation. Cyclic negative-Nyquist conversion power, complete-interpolant error, deliberate-truncation error, and total consequence are reported separately in both normalized Hamiltonian and sorted-band metrics. An independent numerical route reconstructs every authoritative value and the digest-bound visualization data; all 56 comparisons pass with maximum discrepancy $1.42 \times 10^{-14}$. These are provisional finite-control results. They do not establish continuum convergence, material validity, uncertainty quantification, publication acceptance, release, or evidence sealing.

**Keywords:** composite band; gauge alignment; block hopping; Fourier locality; Procrustes problem; independent verification

## 1 Scientific question and evidence boundary

For a fixed isolated rank-two spectral subspace of an exact finite periodic parent, how do a declared reciprocal-space frame attack, pointwise right-unitary alignment, and one-global-unitary alignment change represented Hamiltonians and the locality of their finite block-hopping truncations?

This report records one local, non-retained development calculation with frozen finite controls. It distinguishes four objects: the parent Hamiltonian in its ordered four-state basis; rank-two frames spanning its two lowest states; two-by-two represented Hamiltonian matrices in those frames; and finite-range approximations constructed from complete block Fourier coefficients. Operators are compared only after their state spaces and right actions have been declared.

The result was produced once in memory, admitted from its exact typed bytes, independently reconstructed, and rendered from digest-bound visualization data. No execution directory, sealed evidence package, publication asset inventory, release, continuum result, uncertainty estimate, or material calculation was created. The figures and hashes below identify provisional review bytes only.

## 2 Literature position, motivation, and contribution boundary

Within an isolated group of bands, a reciprocal-space frame may be changed by a momentum-dependent unitary matrix without changing the spectral subspace. Gauge choice, frame smoothness, and real-space localization are central to the construction of maximally localized Wannier functions (MLWFs) for isolated and composite bands [7, 9, 6]. Localized real-space representations also underlie efficient interpolation of reciprocal-space observables [11], and the modern Wannier software ecosystem supports these techniques in material calculations[5]. More generally, the choice of a reduced Hamiltonian representation can matter for observables evaluated after truncation [1].

This report does not propose a new Wannier localization functional, perform spread minimization, or replace disentanglement. It isolates a narrower prerequisite for interpreting reduced Hamiltonians: a pointwise unitary similarity preserves the complete retained spectrum but can redistribute a matrix field among Fourier hopping blocks. Consequently, finite-range truncation is a property of a declared represented operator, frame, transform convention, and support rule; it is not determined by the spectrum alone.

The calculation separates five mathematical objects that are often conflated:

1. the four-state parent operator on $\mathbb{C}^4$;
2. the rank-two spectral subspace as a point of the Grassmannian;
3. a particular orthonormal frame for that subspace;
4. the $2 \times 2$ Hamiltonian represented in that frame; and
5. a finite-support Fourier truncation of that represented field.

A change at one level need not be a change at another. The prescribed attack changes the frame and represented matrix field while preserving the subspace, projector, and complete pointwise spectrum. The later truncation changes the operator and generally its spectrum.

The numerical design draws on classical polar decomposition and orthogonal Procrustes analysis[3, 8, 2], finite trigonometric transforms[10], and standard floating-point caution [4]. Its contribution is not any one of these tools. The contribution is their explicit composition into a typed calculation, independent-verification, and rendering lifecycle that keeps subspace, gauge, transform, and truncation errors separate.

The calculation isolates this mechanism without density-functional theory, WANNIER90, experimental data, or external simulation. It uses an exact synthetic four-state parent and a bounded reciprocal inventory. The construction is not a model of a particular material and does not establish continuum electronic- structure convergence. The one-global-unitary family is included only as a restricted alignment diagnostic. The locality study uses the reference, attacked, and pointwise- aligned channels. Closure eigenphases diagnose the finite transport path; they are not presented as Wilson-loop or topological invariants.

This technical report is intentionally upstream and more complete than publication prose. It records theory, benchmark roles, diagnostics, exact finite results, historical parity, and provenance so that a paper can distill the argument rather than become its only technical authority. Publication allocation and evidence-manifest changes remain separate reviewed actions.

# 3 Exact parent and retained state space

## 3.1 Parent Hamiltonian

Let $k \in [-1/2, 1/2)$ denote reduced reciprocal momentum with period one. The parent Hamiltonian is

$$\mathsf{H}(k) = \mathsf{T}_0 + \mathrm{e}^{2\pi \mathrm{i} k}\mathsf{T}_{+1} + \mathrm{e}^{-2\pi \mathrm{i} k}\mathsf{T}_{+1}^{\dagger}, \qquad \mathsf{H}(k) \in \mathbb{C}^{4\times 4}, \tag{1}$$

with exact rational coefficients in units of $E_{\mathrm{ref}} = 1$,

$$\mathsf{T}_0 = \begin{pmatrix} -6/5 & 3/50 & 0 & 0 \\ 3/50 & -7/20 & 0 & 0 \\ 0 & 0 & 1 & 1/25 \\ 0 & 0 & 1/25 & 7/4 \end{pmatrix}, \qquad \mathsf{T}_{+1} = \begin{pmatrix} 3/20 & 0 & 3/25 & 0 \\ 0 & -1/10 & 0 & 1/10 \\ 0 & 0 & 1/20 & 0 \\ 0 & 0 & 0 & -1/25 \end{pmatrix}. \tag{2}$$

The ordered four-state coefficient basis, onsite energy zero, reciprocal period, and energy reference are part of the typed configuration. No fitted or historical campaign value is read at runtime.

Let

$$\lambda_1(k) \le \lambda_2(k) \le \lambda_3(k) \le \lambda_4(k) \tag{3}$$

be the ordered eigenvalues of $\mathsf{H}(k)$. The retained spectral space is the span of the first two eigenvectors. The sampled retained/excluded gap is

$$\Delta_{23}^{\mathrm{sampled}} = \min_{k\in\mathcal{K}_{64}} \big[\lambda_3(k) - \lambda_2(k)\big]. \tag{4}$$

This is a finite-mesh qualification diagnostic, not a proof of a global gap over the continuous Brillouin zone.

## 3.2 Construction and evaluation inventories

The centered construction mesh is

$$\mathcal{K}_{64} = \left\{ -\frac{1}{2} + \frac{j}{64} : j = 0, \ldots, 63 \right\}. \tag{5}$$

It constructs frames, alignments, represented operators, complete Fourier blocks, and truncated models. The evaluation mesh is

$$\mathcal{Q}_{257} = \left\{ -\frac{1}{2} + \frac{j + 1/65}{257} : j = 0, \ldots, 256 \right\}. \tag{6}$$

The two inventories are proven disjoint using exact rational arithmetic before their conversion to binary64. Evaluation values cannot modify the construction.

# 4 Frames, attack, and right-unitary alignment

## 4.1 Reference frame transport

At every construction point, the dense complex-Hermitian eigensystem supplies a raw frame $\mathsf{F}_{\mathrm{raw}}(k) \in \mathbb{C}^{4\times 2}$ whose columns are the two lowest ordered eigenvectors. Neighboring frames are aligned by the unitary polar factor of their overlap, following the standard polar-decomposition construction [3]. The selected closure transformation is distributed around the finite reciprocal cycle, producing the reference path $\mathsf{F}_{\mathrm{ref}}(k)$. For every point,

$$\mathsf{F}_{\mathrm{ref}}(k)^{\dagger}\mathsf{F}_{\mathrm{ref}}(k) = \mathsf{I}_2. \tag{7}$$

The smallest singular value among neighbor and closure overlaps is retained as a transport-conditioning diagnostic.

### 4.2 Declared attack

The reference path is rotated on the right by

$$\mathsf{A}(k) = \begin{pmatrix} \cos\theta(k) & -\sin\theta(k) \\ \sin\theta(k) & \cos\theta(k) \end{pmatrix}, \qquad \theta(k) = 0.30 + 0.65\sin(2\pi k) + 0.30\sin(4\pi k), \tag{8}$$

so that

$$\mathsf{F}_{\mathrm{att}}(k) = \mathsf{F}_{\mathrm{ref}}(k)\mathsf{A}(k). \tag{9}$$

The attack preserves the retained projector exactly in exact arithmetic but changes coordinates within the retained space.

### 4.3 Pointwise and global feasible families

For each construction point, pointwise alignment solves the right-action orthogonal Procrustes problem

$$\mathsf{Q}_k = \arg\min_{\mathsf{Q}\in U(2)} \|\mathsf{F}_{\mathrm{att}}(k)\mathsf{Q} - \mathsf{F}_{\mathrm{ref}}(k)\|_{\mathrm{F}}\,. \tag{10}$$

The orthogonal Procrustes problem and its singular-value solution are classical [8, 2]. A separate calculation selects one common unitary

$$\mathsf{Q}_{\mathrm{global}} = \arg\min_{\mathsf{Q}\in U(2)} \sum_{k\in\mathcal{K}_{64}} \|\mathsf{F}_{\mathrm{att}}(k)\mathsf{Q} - \mathsf{F}_{\mathrm{ref}}(k)\|_{\mathrm{F}}^2\,. \tag{11}$$

These are different feasible sets; neither approximates the other by definition.

For both families the rank cutoff is

$$\tau_{\mathrm{rank}} = \max\left(10^{-14}, 10^{-12}\sigma_{\max}\right). \tag{12}$$

Both covariance singular values must exceed this cutoff. Rank deficiency fails closed, with no null-space completion.

## 5 Represented operators and block Fourier locality

For a frame $\mathsf{F}_c(k)$ in channel $c$, the represented Hamiltonian is

$$\mathsf{h}_c(k) = \mathsf{F}_c(k)^{\dagger}\mathsf{H}(k)\mathsf{F}_c(k) \in \mathbb{C}^{2\times 2}. \tag{13}$$

The reference, attacked, pointwise-aligned, and globally aligned paths are retained. Only the first three enter the Fourier-locality study.

With centered representatives $r = -32, \ldots, 31$, the complete cyclic transform pair is

$$\mathsf{t}_{c,r}^{\mathrm{cyc}} = \frac{1}{64}\sum_{k\in\mathcal{K}_{64}} \mathrm{e}^{-2\pi\mathrm{i}rk}\mathsf{h}_c(k), \tag{14}$$

$$\mathsf{h}_c(k_j) = \sum_{r=-32}^{31} \mathrm{e}^{2\pi\mathrm{i}rk_j}\mathsf{t}_{c,r}^{\mathrm{cyc}}. \tag{15}$$

The cyclic transform retains all 64 blocks and checks both construction reconstruction and cyclic block Hermiticity. Before applying an ordinary-lattice range cutoff, the single negative-Nyquist representative is removed. The resulting support is $\mathcal{R}_{\mathrm{ord}} = \{-31, \ldots, 31\}$, with conversion power

$$P_{c,\mathrm{Nyq}} = \frac{\|\mathsf{t}_{c,-32}^{\mathrm{cyc}}\|_{\mathrm{F}}^2}{2E_{\mathrm{ref}}^2}. \tag{16}$$

This is a normalized block-power diagnostic, not an off-node spectral error. The complete ordinary interpolant and its symmetric range truncation are

$$\mathsf{h}_c^{\mathrm{ord}}(q) = \sum_{r=-31}^{31} \mathrm{e}^{2\pi \mathrm{i} r q} \mathsf{t}_{c,r}^{\mathrm{cyc}}, \tag{17}$$

$$\mathsf{h}_c^{(R)}(q) = \sum_{|r|\le R} \mathrm{e}^{2\pi \mathrm{i} r q} \mathsf{t}_{c,r}^{\mathrm{cyc}}, \qquad R \in \{0,1,2,3,4,6,8\}. \tag{18}$$

The ordinary omitted-block norm is

$$d_{c,\mathrm{ord}}^{(R)} = \left( \sum_{r \in \mathcal{R}_{\mathrm{ord}},\, |r|>R} \|\mathsf{t}_{c,r}^{\mathrm{cyc}}\|_{\mathrm{F}}^2 \right)^{1/2}. \tag{19}$$

On the disjoint evaluation mesh, the direct reference and pointwise targets are $\mathsf{h}_{\mathrm{R}}^{\mathrm{ord}}(q)$. The direct attacked target applies the analytic rotation without re-transforming attacked samples,

$$\mathsf{h}_{\mathrm{A}}^{\mathrm{dir}}(q) = \mathsf{A}(q)^{\dagger} \mathsf{h}_{\mathrm{R}}^{\mathrm{ord}}(q) \mathsf{A}(q). \tag{20}$$

For compatible matrix fields define

$$\delta_{H,\max}(X,Y) = \max_{q \in \mathcal{Q}_{257}} \frac{\|X(q) - Y(q)\|_{\mathrm{F}}}{\sqrt{2} E_{\mathrm{ref}}}, \tag{21}$$

$$\delta_{E,\max}(X,Y) = \max_{q \in \mathcal{Q}_{257}} \max_{n=1,2} \frac{|\lambda_n(X(q)) - \lambda_n(Y(q))|}{E_{\mathrm{ref}}}. \tag{22}$$

Both metrics are reported in three distinct channels:

$$\Delta_{c,\mathrm{interp}} : \mathsf{h}_c^{\mathrm{ord}} - \mathsf{h}_c^{\mathrm{dir}}, \tag{23}$$

$$\Delta_{c,\mathrm{trunc}} : \mathsf{h}_c^{(R)} - \mathsf{h}_c^{\mathrm{ord}}, \tag{24}$$

$$\Delta_{c,\mathrm{total}} : \mathsf{h}_c^{(R)} - \mathsf{h}_c^{\mathrm{dir}}. \tag{25}$$

The construction total spectral error continues to compare the truncated spectrum with the exact retained parent spectrum on $\mathcal{K}_{64}$. The three disjoint-mesh maxima are not added as independent errors; their underlying pointwise matrix residuals obey a triangle inequality and may reinforce or cancel. This decomposition prevents finite interpolation from being misreported as deliberate support deletion. Spectral equality at each untruncated reciprocal point does not imply equal blocks or equal finite-range errors.

# 6 Exact analytic consequences

## 6.1 Projector and spectral invariance under the attack

**Proposition 1** (Pointwise equivalence)**.** *Let* $\mathsf{F}(k) \in \mathbb{C}^{4\times 2}$ *have orthonormal columns, let* $\mathsf{A}(k) \in U(2)$*, and define* $\mathsf{F}_{\mathrm{A}}(k) = \mathsf{F}(k)\mathsf{A}(k)$*. Then*

$$\mathsf{F}_{\mathrm{A}}(k)\mathsf{F}_{\mathrm{A}}(k)^{\dagger} = \mathsf{F}(k)\mathsf{F}(k)^{\dagger}, \tag{26}$$

$$\mathsf{h}_{\mathrm{A}}(k) = \mathsf{A}(k)^{\dagger} \mathsf{h}_{\mathrm{R}}(k) \mathsf{A}(k). \tag{27}$$

*The attacked and reference represented operators therefore have identical ordered eigenvalues at every momentum.*

*Proof.* The projector identity follows from $\mathsf{A}\mathsf{A}^\dagger = \mathsf{I}_2$. Substituting $\mathsf{F}_\mathrm{A} = \mathsf{F}\mathsf{A}$ into Equation (13) gives Equation (27). Unitary similarity preserves the characteristic polynomial and hence the ordered spectrum. □

**Corollary 1** (Exact pointwise recovery)**.** *The right action $\mathsf{Q}_k = \mathsf{A}(k)^\dagger$ maps the attacked frame exactly to the reference frame and maps the attacked represented operator exactly to the reference represented operator in exact arithmetic.*

This corollary makes the pointwise route an analytic oracle rather than merely another numerical fit. Failure to recover the reference within tolerance would identify an implementation, conditioning, or convention error.

### 6.2 Why one global unitary cannot invert the declared attack

Suppose one constant $\mathsf{Q}$ satisfied $\mathsf{F}_\mathrm{att}(k)\mathsf{Q} = \mathsf{F}_\mathrm{ref}(k)$ for every construction point. Left multiplication by $\mathsf{F}_\mathrm{ref}(k)^\dagger$ would imply $\mathsf{A}(k)\mathsf{Q} = \mathsf{I}_2$, hence $\mathsf{Q} = \mathsf{A}(k)^\dagger$ for every $k$. The attack angle in Equation (8) is not constant on $\mathcal{K}_{64}$, so no such common inverse exists. The global Procrustes result is therefore a best restricted approximation, not an expected recovery channel.

### 6.3 Fourier convolution under momentum-dependent similarity

Write compatible Fourier expansions

$$\mathsf{h}_\mathrm{R}(k) = \sum_r \mathsf{t}_r \mathrm{e}^{2\pi \mathrm{i} k r}, \qquad \mathsf{A}(k) = \sum_s \mathsf{A}_s \mathrm{e}^{2\pi \mathrm{i} k s}. \tag{28}$$

Substitution into Equation (27) gives attacked blocks

$$\mathsf{t}^\mathrm{A}_n = \sum_{p,q} \mathsf{A}_p^\dagger \mathsf{t}_{n+p-q} \mathsf{A}_q. \tag{29}$$

A constant unitary has only $\mathsf{A}_0 \neq 0$, so $\mathsf{t}^\mathrm{A}_n = \mathsf{A}_0^\dagger \mathsf{t}_n \mathsf{A}_0$; support and block Frobenius norms are unchanged. A momentum-dependent unitary has multiple Fourier components, and Equation (29) mixes hopping indices. This is the exact mechanism by which complete pointwise spectral equivalence can coexist with a longer represented block tail.

### 6.4 Truncation does not commute with the attack

Let $\mathcal{T}_R$ denote projection onto the ordinary support $|r| \leq R$. In general,

$$\mathcal{T}_R[\mathsf{A}^\dagger \mathsf{h} \mathsf{A}] \neq \mathsf{A}^\dagger \mathcal{T}_R[\mathsf{h}]\mathsf{A}, \tag{30}$$

because the right side still contains the full Fourier content of $\mathsf{A}(k)$. The truncated reference and attacked fields therefore need not remain pointwise unitarily similar. Their post-truncation spectra may differ even though their complete spectra agree exactly.

The complete-transform closure check is essential here. If the full transform failed to reconstruct the represented field, one could not attribute the observed difference to deliberate truncation. Closure first, truncation second, and disjoint evaluation third are separate protocol stages.

### 6.5 Block Parseval identity and spectral bound

For the complete finite transform, matrix-valued Parseval orthogonality gives

$$\frac{1}{64}\sum_{k\in\mathcal{K}_{64}}\left\|\mathsf{h}_c^{\mathrm{ord}}(k)-\mathsf{h}_c^{(R)}(k)\right\|_{\mathrm{F}}^2=\sum_{r\in\mathcal{R}_{\mathrm{ord}},\,|r|>R}\|\mathsf{t}_{c,r}^{\mathrm{cyc}}\|_{\mathrm{F}}^2=\left(d_{c,\mathrm{ord}}^{(R)}\right)^2. \tag{31}$$

For each sampled momentum, the ordered Hermitian eigenvalue perturbation is bounded by the spectral norm and hence by the Frobenius norm,

$$\max_n|\lambda_n(\mathsf{h}_c^{(R)})-\lambda_n(\mathsf{h}_c^{\mathrm{ord}})|\le\|\mathsf{h}_c^{(R)}-\mathsf{h}_c^{\mathrm{ord}}\|_2\le\|\mathsf{h}_c^{(R)}-\mathsf{h}_c^{\mathrm{ord}}\|_{\mathrm{F}}. \tag{32}$$

Equations (31) and (32) are verification constraints. They do not produce a universal continuum tail-to-band-error law.

## 7 Methods and benchmark matrix

### 7.1 Ordered calculation protocol

The calculation executes the following ordered stages:

1. construct exact-rational parent blocks and evaluate Hermitian fibers;
2. compute ordered eigensystems and qualify the sampled retained/excluded gap;
3. polar-transport the retained frame and qualify every overlap;
4. apply the declared right-unitary attack;
5. solve pointwise and one-global-unitary Procrustes problems with fail-closed rank selection;
6. construct represented operators and compare frames, projectors, right actions, and operators separately;
7. form complete centered cyclic block transforms and require reconstruction and cyclic Hermiticity closure;
8. remove and separately measure the negative-Nyquist block before constructing complete ordinary-lattice interpolants;
9. construct direct reference, attacked, and pointwise targets on the disjoint mesh;
10. truncate the three ordinary channels on seven symmetric supports;
11. evaluate interpolation, deliberate-truncation, and total errors in normalized Hamiltonian and sorted-band metrics, while retaining construction total errors; and
12. independently reconstruct every authoritative field and derive the visualization data used by the report figures.

A failed gate terminates the scientific route; downstream values are not interpreted when their prerequisites fail.

Table 1: Benchmark roles in the rank-two control.

| Benchmark | Question | Decisive diagnostic |
|---|---|---|
| B0 parent | Is the retained space sampled-separated? | Minimum second-to-third gap versus $0.5E_{\mathrm{ref}}$ |
| B1 transport | Is frame propagation numerically conditioned? | Minimum overlap singular value versus 0.8 |
| B2 attack | Does the declared right action change only coordinates? | Projector equality and exact pointwise spectra |
| B3 pointwise oracle | Is the known inverse recovered? | Frame, projector, action, and operator defects |
| B4 global restriction | Can one constant unitary undo the attack? | Global frame and operator residuals |
| B5 transform | Are locality differences preceded by complete closure? | Reconstruction, cyclic Hermiticity, and conversion power |
| B6 error channels | What changes at interpolation and support deletion? | Separate interpolation, truncation, and total Hamiltonian and spectral maxima |

### 7.2 Seven complementary benchmarks

Historical M2 parity is an additional provenance benchmark, not a runtime dependency. It tests named finite channels against predecessor values while allowing explicitly documented changes in canonical transport and centered-transform conventions.

### 7.3 Numerical ownership and failure policy

PhysKit owns dense complex-Hermitian eigensystems. KSD owns the parent semantics, transport composition, represented operators, block transforms, result records, and scientific decisions. Project Koios Optimization owns the unit-free batched right-action Procrustes selection. The independent verifier bypasses the production analyzer and the Optimization Procrustes solver.

Rank selection uses Equation (12). At or below the selected cutoff, the calculation fails rather than completing an arbitrary null space. Construction and evaluation coordinates are proven disjoint in exact rational arithmetic. Binary64 is used only after this combinatorial property is established.

## 8 Finite numerical results

### 8.1 Qualification and alignment diagnostics

The minimum sampled retained/excluded gap was

$$\Delta_{23}^{\mathrm{sampled}} = 1.0566461211497054\, E_{\mathrm{ref}}, \tag{33}$$

which exceeds the declared threshold $0.5E_{\mathrm{ref}}$. The minimum neighbor/closure overlap singular value was 0.9999828868735373, above the declared threshold 0.8. The two closure eigenphases were $-0.018778493443278994$ and $-0.014159454102255378$ radians. These are bounded transport diagnostics only.

Table 2 separates the frame, projector, and represented-operator channels. The attack changes both frame coordinates and the represented matrix field. Pointwise alignment recovers the reference at roundoff, including the known right transformation. One global unitary reduces neither defect to roundoff because it cannot represent the momentum-dependent attack.

Table 2: Finite alignment diagnostics. Operator defects are reported in units of $E_{\mathrm{ref}}$; frame, projector, and rotation defects are dimensionless Frobenius norms.

| Diagnostic | Value |
|---|---|
| Attack frame maximum defect | 1.5056428951173255 |
| Pointwise frame maximum defect | $5.161779512921633 \times 10^{-16}$ |
| Pointwise projector maximum defect | $3.603691565314897 \times 10^{-16}$ |
| Pointwise attack-recovery maximum defect | $6.600445091476726 \times 10^{-16}$ |
| One-global-unitary frame maximum defect | 1.1309249755520259 |
| Attacked operator maximum defect | $0.9972076810972789\, E_{\mathrm{ref}}$ |
| Pointwise operator maximum defect | $9.310752561074258 \times 10^{-16}\, E_{\mathrm{ref}}$ |
| One-global-unitary operator maximum defect | $0.7442844384409564\, E_{\mathrm{ref}}$ |
| Pointwise minimum covariance singular value | 0.9999999999999984 |
| Pointwise maximum applied rank cutoff | $1.0000000000000028 \times 10^{-12}$ |
| Global aggregate minimum covariance singular value | 56.132454593539926 |
| Global aggregate applied rank cutoff | $5.613245459353998 \times 10^{-11}$ |

The aggregate global covariance sums 64 ordered contributions, so its singular values have a different scale from the pointwise covariances. In both cases the smaller singular value is far above the applicable maximum-rule cutoff; no rank completion is used.

### 8.2 Complete-transform closure

Table 3: Complete 64-block transform diagnostics. Reconstruction and cyclic Hermiticity maxima are in units of $E_{\mathrm{ref}}$; conversion power is normalized by $2E_{\mathrm{ref}}^2$.

| Channel | Reconstruction maximum | Cyclic Hermiticity maximum | $P_{\mathrm{Nyq}}$ |
|---|---|---|---|
| Reference | $7.0253 \times 10^{-16}$ | $9.3248 \times 10^{-18}$ | $6.6400 \times 10^{-33}$ |
| Attacked | $1.0775 \times 10^{-15}$ | $3.9303 \times 10^{-17}$ | $2.5042 \times 10^{-33}$ |
| Pointwise | $9.0445 \times 10^{-16}$ | $3.1940 \times 10^{-17}$ | $4.5767 \times 10^{-34}$ |

All three complete transforms reconstruct their construction matrix fields at roundoff. Their removed cyclic Nyquist powers are below $7 \times 10^{-33}$ and remain separate from ordinary omitted-tail accounting. This closure and conversion sequence precedes every truncation and prevents range error from being confused with an incomplete transform or cyclic-to-ordinary convention change.

### 8.3 Range dependence and gauge-sensitive locality

Figure 1 shows the two retained parent bands and the principal frame and represented-operator defects. Pointwise defects lie near roundoff, whereas the attack and global-family defects remain order one on the dimensionless reference scale.

Figure 2 reorganizes the same six defects by operation. The attacked and globally aligned channels retain order-one frame and represented-operator defects, while the pointwise channel reaches binary64-scale residuals in both measures. Pairing the two diagnostics prevents a small operator residual from being read as frame recovery, or conversely.

The complete range inventory is reported in Table 4. Reference and pointwise results agree to roundoff throughout. Their total evaluation errors decrease from approximately $2.98 \times 10^{-1} E_{\mathrm{ref}}$ at

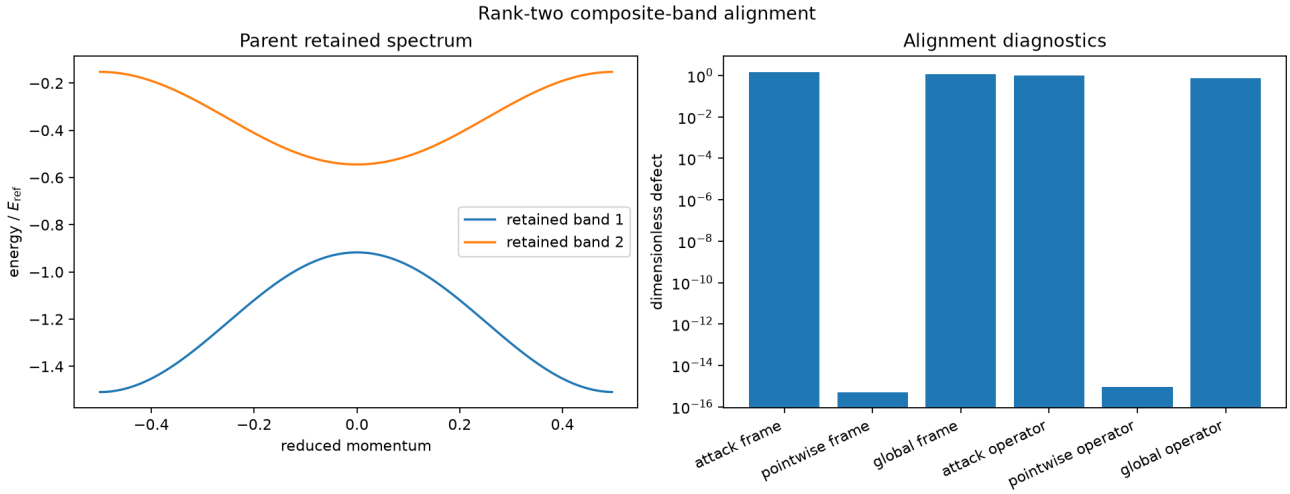



Figure 1: Finite-mesh alignment diagnostics. The left panel shows the two retained parent eigenvalues on the disjoint evaluation mesh. The right panel reports frame and represented-operator defects for the attacked, pointwise-aligned, and globally aligned channels. The common logarithmic scale preserves the distinction between these quantities; it does not identify them.

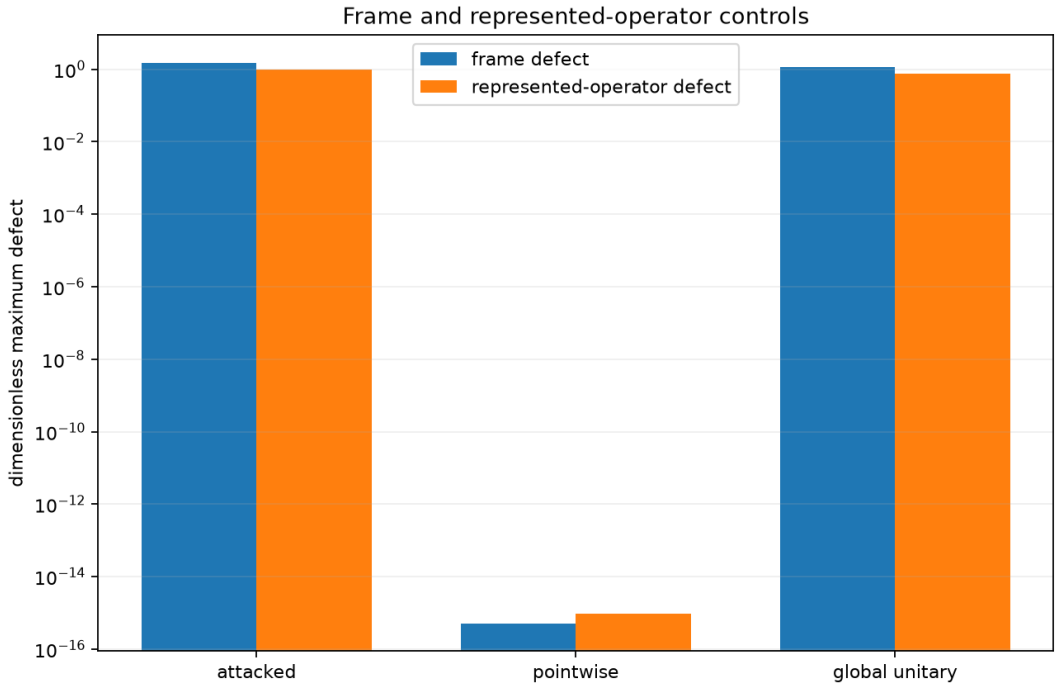



Figure 2: Frame and represented-operator controls for the three non-reference channels. Each alignment family is shown with separate maximum frame and operator defects. The paired bars answer different questions and are not combined into a single alignment score.

onsite range to $1.17 \times 10^{-14} E_{\mathrm{ref}}$ at range eight. The attacked error decreases much more slowly and remains $7.041355133552418 \times 10^{-5} E_{\mathrm{ref}}$ at range eight, approximately $6.2 \times 10^{9}$ times the reference value there. This ratio is a finite sampled comparison, not an asymptotic decay rate.

Table 4: Complete finite range study. Ordinary omitted norms $d$, construction total errors $e_{\mathrm{con}}$, and disjoint-evaluation total errors $e_{\mathrm{eval}}$ are in units of $E_{\mathrm{ref}}$. R, A, and P denote reference, attacked, and pointwise channels.

| | Ordinary omitted block $L^2$ norm | | | Construction total spectral maximum | | | Evaluation total spectral maximum | | |
|---|---|---|---|---|---|---|---|---|---|
| $R$ | $d_{\mathrm{R}}$ | $d_{\mathrm{A}}$ | $d_{\mathrm{P}}$ | $e_{\mathrm{R,con}}$ | $e_{\mathrm{A,con}}$ | $e_{\mathrm{P,con}}$ | $e_{\mathrm{R,eval}}$ | $e_{\mathrm{A,eval}}$ | $e_{\mathrm{P,eval}}$ |
| 0 | 2.5450e−1 | 5.2350e−1 | 2.5450e−1 | 2.9789e−1 | 4.4454e−1 | 2.9789e−1 | 2.9789e−1 | 4.4454e−1 | 2.9789e−1 |
| 1 | 1.9754e−5 | 1.8512e−1 | 1.9754e−5 | 2.7573e−5 | 1.4768e−1 | 2.7573e−5 | 2.7565e−5 | 1.4898e−1 | 2.7565e−5 |
| 2 | 8.7006e−7 | 9.7439e−2 | 8.7006e−7 | 1.2269e−6 | 9.7312e−2 | 1.2269e−6 | 1.2261e−6 | 9.7379e−2 | 1.2261e−6 |
| 3 | 3.8719e−8 | 3.9152e−2 | 3.8719e−8 | 5.5322e−8 | 4.1515e−2 | 5.5322e−8 | 5.5259e−8 | 4.1531e−2 | 5.5259e−8 |
| 4 | 1.7453e−9 | 8.2797e−3 | 1.7453e−9 | 2.4856e−9 | 6.3755e−3 | 2.4856e−9 | 2.4869e−9 | 6.4296e−3 | 2.4869e−9 |
| 6 | 3.8610e−12 | 5.2560e−4 | 3.8609e−12 | 5.0475e−12 | 5.5446e−4 | 5.0472e−12 | 5.0389e−12 | 5.5899e−4 | 5.0389e−12 |
| 8 | 1.6352e−14 | 5.6539e−5 | 1.6370e−14 | 1.1435e−14 | 6.9238e−5 | 1.1324e−14 | 1.1657e−14 | 7.0414e−5 | 1.1546e−14 |

Figure 3 presents the omitted-block norms as a diagnostic cutoff sweep. The coefficients are measured without modification: selecting a point on this curve does not yet truncate the represented field. Reference and pointwise values remain visually coincident and decay to approximately $1.6 \times 10^{-14} E_{\mathrm{ref}}$ at $S = 8$, whereas the attacked tail remains $5.65 \times 10^{-5} E_{\mathrm{ref}}$. The common integer ranges are also used by the retained-support analysis, but the diagnostic and model-reduction operations remain distinct.

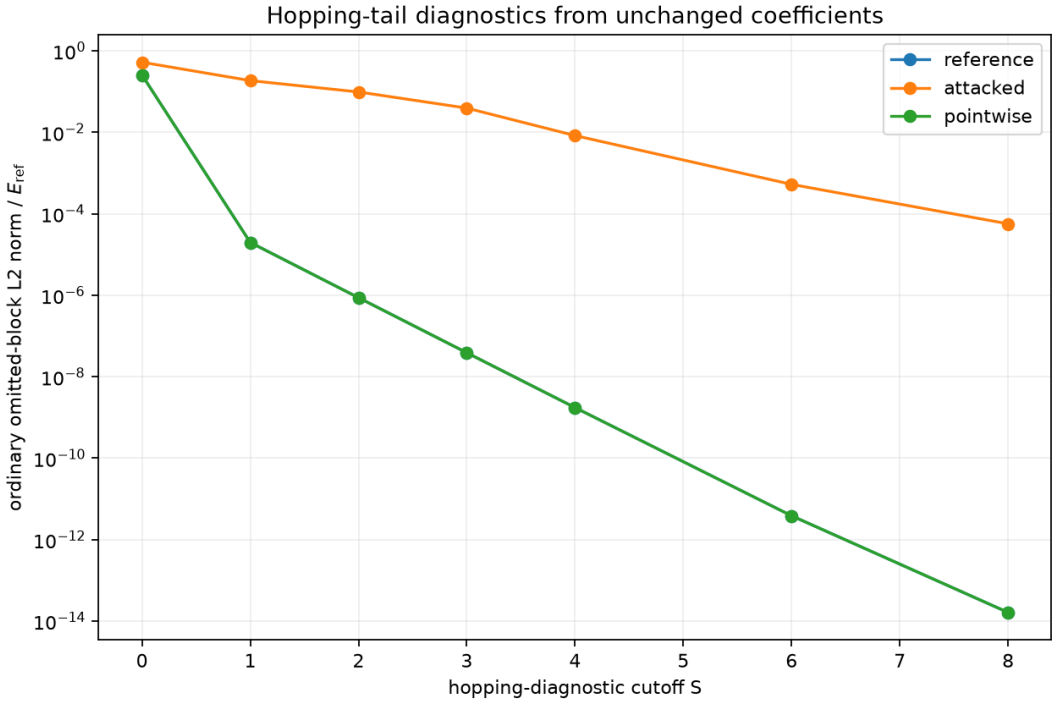



Figure 3: Hopping-tail diagnostics computed from unchanged ordinary Fourier coefficients. The abscissa is a diagnostic cutoff, and the ordinate is the omitted-block $L^2$ norm. The curves measure locality; they do not apply a support-truncation policy.

Figure 4 shows both the spectral consequence and the underlying complete-block decay. The reference and pointwise curves are visually coincident because pointwise alignment recovers the reference represented operator. The attacked channel redistributes matrix weight over substantially more

representatives. Exact pointwise band equivalence is therefore compatible with very different finite-range locality.

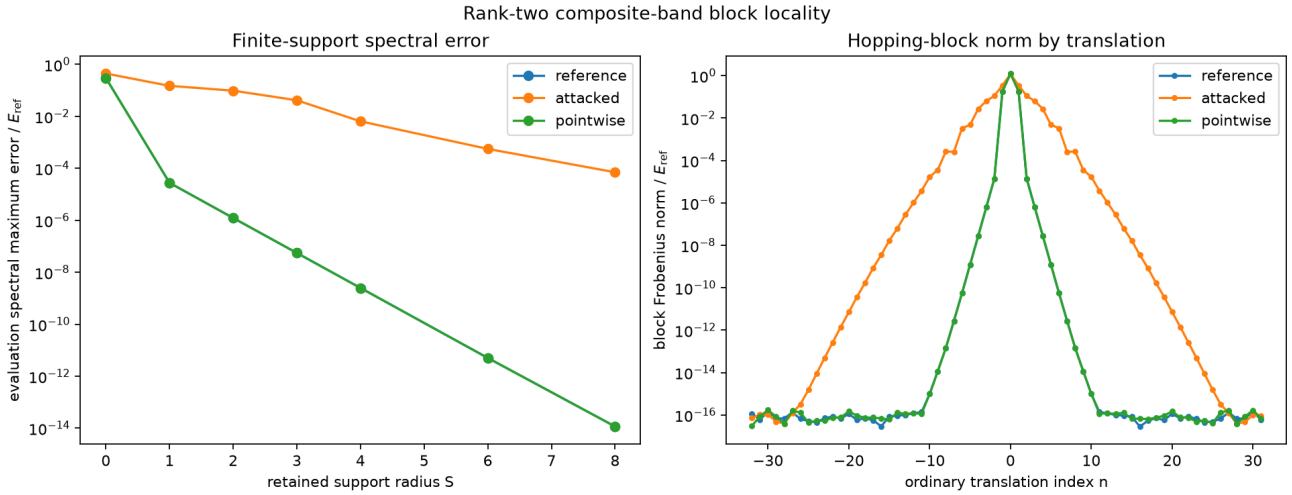



Figure 4: Finite-support error and complete hopping-block locality. The left panel reports the maximum total ordered-spectrum error on the disjoint evaluation inventory for each retained support. The right panel reports the Frobenius norm of every block in the complete 64-point ordinary-translation inventory. Coincident reference and pointwise curves reflect represented-operator recovery, whereas the extended attacked tail exhibits the frame dependence of the block distribution.

### 8.4 Separated error-channel accounting

The complete ordinary reference interpolant is its own direct target, so its interpolation residual is exactly zero under the implemented field definition. For the attacked channel, the maximum interpolation residual is $1.524695274864153 \times 10^{-15}$ in normalized Frobenius error and $1.998401444325282 \times 10^{-15}$ in sorted-band error. The pointwise values are $1.104000291768802 \times 10^{-15}$ and $1.554312234475219 \times 10^{-15}$, respectively. Thus finite interpolation remains at binary64 scale and does not explain the displayed attacked range trend.

At range four, the attacked deliberate-truncation and total normalized errors are

$$\delta_{H,\mathrm{trunc}} = 9.142140906351088 \times 10^{-3}, \qquad \delta_{H,\mathrm{total}} = 9.142140906351100 \times 10^{-3}, \tag{34}$$

$$\delta_{E,\mathrm{trunc}} = 6.429588433655875 \times 10^{-3}, \qquad \delta_{E,\mathrm{total}} = 6.429588433655653 \times 10^{-3}. \tag{35}$$

At range eight, the attacked truncation and total sorted-band errors are $7.041355133541316 \times 10^{-5}$ and $7.041355133552418 \times 10^{-5}$, respectively. Their near equality is an observed consequence of the roundoff-scale interpolation channel, not an assumption that maxima add.

Figure 6 extends the decomposition to every locality channel. For each retained support, interpolation compares the complete interpolant with the direct target, truncation compares the retained-support field with the complete interpolant, and total error compares the retained-support field with the direct target. Reference and pointwise errors decay rapidly with support, whereas the attacked channel retains substantially larger truncation and total errors. Values equal to zero are placed at $10^{-18}$ for logarithmic display only; the retained numerical values are not altered.

No application-error threshold or preferred hopping range is declared by this protocol. Consequently the calculation supports a comparison of channels, not a range-selection or admissibility claim.

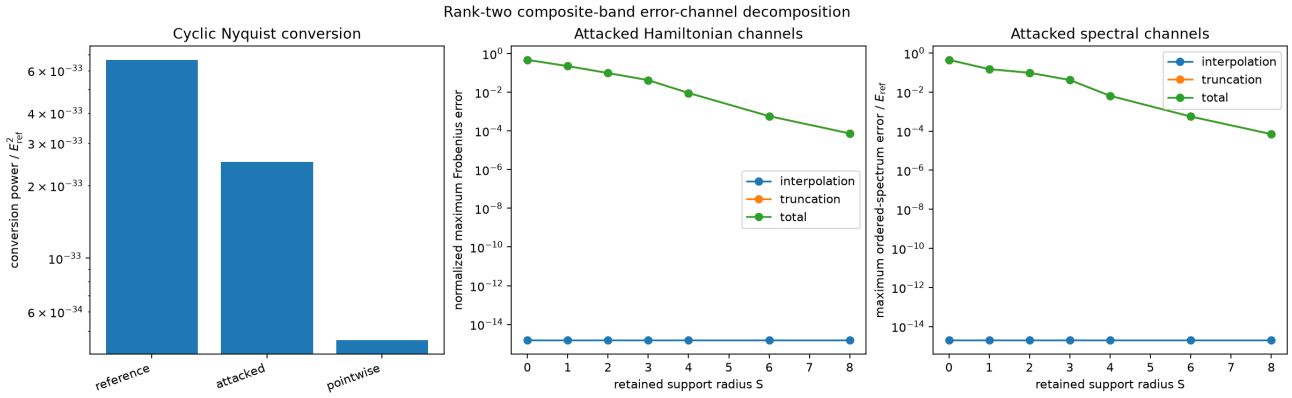



Figure 5: Separated error-channel accounting. The left panel reports normalized negative-Nyquist conversion power independently of off-node errors. The center and right panels show interpolation, deliberate-truncation, and total maximum residuals for the attacked channel in normalized Hamiltonian and ordered-spectrum metrics. Lines connect the seven retained supports only as visual guides; the channels are neither stacked nor summed.

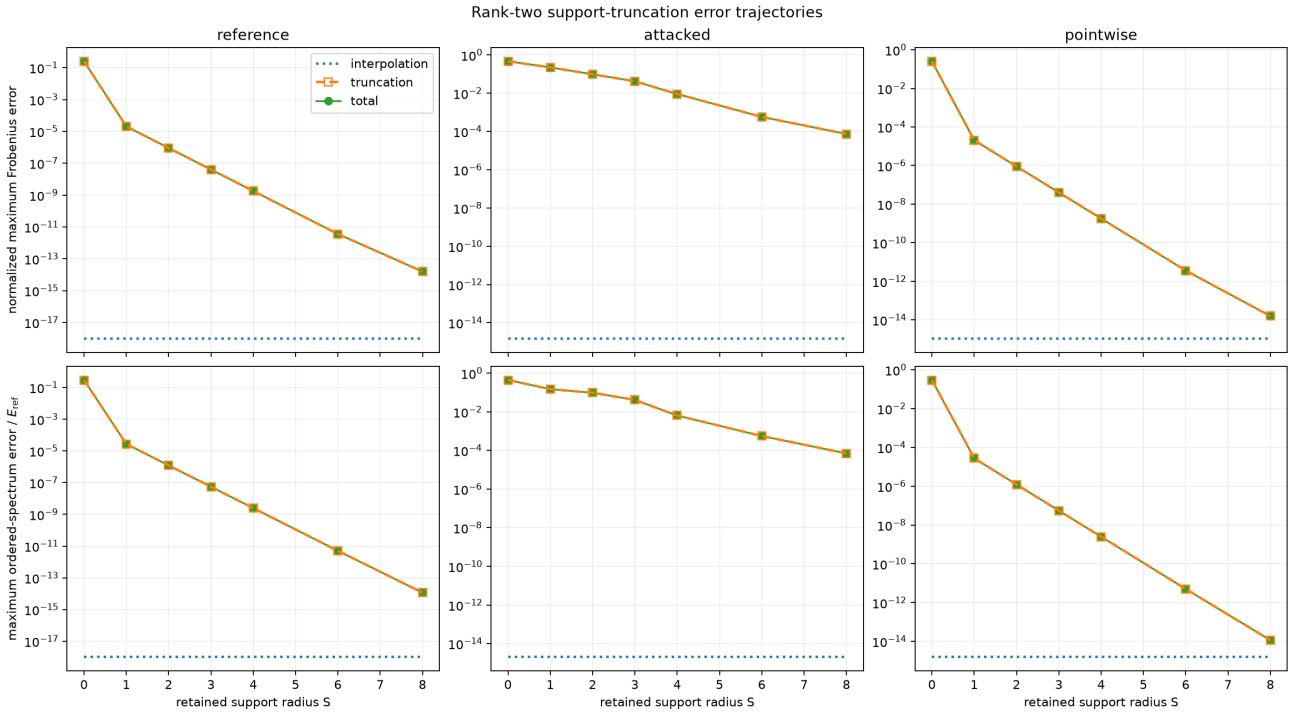



Figure 6: Non-stacked support-truncation error trajectories for the reference, attacked, and pointwise channels. The upper row reports normalized operator error and the lower row ordered-spectrum error. Interpolation, deliberate truncation, and total error remain separate channels even where their plotted curves coincide.

## 9 Independent verification and figure provenance

The producer uses PhysKit dense complex-Hermitian eigensystems, the `ksdft2effmass` (KSD) polar frame-transport and represented-operator owners, and the Project Koios Optimization batched complex-unitary Procrustes solver. The independent verifier does not call the producer or that Procrustes solver. It reconstructs the exact parent, canonical eigensystem phases, frame transport, attacks, singular-value right actions, represented operators, complete cyclic and ordinary Fourier sums, negative-Nyquist conversion power, direct evaluation targets, block truncations, separated Hamiltonian and ordered-spectrum error channels, diagnostics, and every field of the visualization data using direct NumPy and SciPy formulas.

The verifier independently applies Equation (12) and rejects any pointwise or global covariance at or below the selected cutoff. Shared binary64, Basic Linear Algebra Subprograms (BLAS), Linear Algebra PACKage (LAPACK), NumPy, and SciPy lineage remains a disclosed limitation; this is an independent software route rather than an independent physical experiment.

All 56 named comparisons passed. The maximum observed discrepancy was $1.4210854715202004 \times 10^{-14}$, below the applicable stored tolerances. The result, verification aggregate, and visualization data each survive canonical byte-identical JavaScript Object Notation (JSON) round trips. Rendering admits exact stored bytes only after recomputing and exactly comparing both verification and visualization values.

The six Portable Network Graphics (PNG) files embedded in Figures 1–6 were produced from the verified, digest-bound visualization data at 160 dots per inch. They contain no newly calculated scientific value and are report-local review figures rather than retained execution artifacts.

## 10 Historical parity and convention boundary

The direct source is the bounded scientific successor to the historical M2 calculation, but it does not decode or wrap historical campaign schemas. Named finite channels reproduce historical M2 values within the direct $10^{-11}$ energy tolerance, including the sampled gap, overlap conditioning, frame and operator defects, omitted-block norms, and construction spectral errors. For example, the historical and direct attacked construction errors at range four are respectively $0.006375530149533293 E_{\mathrm{ref}}$ and $0.006375530149533404 E_{\mathrm{ref}}$.

The direct lifecycle uses current canonical KSD transport and centered Fourier owners. It therefore does not assert byte-identical closure phases, hopping blocks, or every attacked evaluation value. At range four, historical M2 reports an attacked evaluation error of $0.006429434409208734 E_{\mathrm{ref}}$, while the direct source reports $0.006429588433655653 E_{\mathrm{ref}}$. This explicit adaptation prevents a compatibility transformation from hiding the current convention. Historical publication claims remain bound to their original result and verification bytes.

## 11 Provisional provenance

The in-memory development calculation used branch `work/periodic1d-composite-alignment-v3` at base revision `8cf9e93d7c3d791f81ee119476328970fe5632ba` in a dirty worktree. The local software environment reported Python 3.14.6, ksdft2effmass 0.1.0.dev0, projectkoios-physkit 0.1.3.dev0, projectkoios-optimization 0.1.0.dev0, NumPy 2.5.1, SciPy 1.18.0, and Matplotlib 3.11.1. These versions describe the review environment; they do not create a retained execution contract.

Exact 256-bit Secure Hash Algorithm (SHA-256) identities for the provisional review bytes are:

- configuration bytes: `08ad9c6d967616e5b286ece1a3296b2ed65bdce672daacaecf26b933e04ca4b6`;

- result bytes: `7c8a65f14f480c22e4eeeeae26f56ef7f0116accec0a16f80bb7c0cf7931f0f7`;
- verification bytes: `72d65f4c67ec0a76b23b333317afbc16736980e877ba0f6f0dffcbf0e56d5102`;
- visualization-data bytes: `578d8270188d25087091862bdb276ff818975f18df051b5a9615ff6f6a5fb1fa`;
- alignment report PNG: `b276f31a8a316102652c496cd06cb5d7ab1f8e51a661eff4f06ad8ed47610368`;
- recovery-control report PNG: `5bd798e781667f602a499a869208ac6ad33c2008bd25f0318fbfbdbd3dfd7451`;
- hopping-diagnostic report PNG: `ac05b560b71bb2433764d02a909a03c47c0d597ccd97f2234c7290ddbe4bdada`;
- locality report PNG: `bb1a8f1ce2fa78674363bde92af6a0b949e62fbbb27d73791bff08298352eb12`;
- error-channel report PNG: `f17402c56310b1982c61ed292bb822bc6057e23f683e2d293e61ed24fc11c2e4`; and
- error-trajectory report PNG: `b94cb9424585729c3fae45aa2c7c87dbf225a7c9373903999ccfc38ca0cbec70`.

The canonical result, verification, and visualization documents contained 961263, 25618, and 45390 bytes respectively. SHA-256 establishes byte identity only; it does not establish scientific validity, authorship, rights, publication, release, or sealing. No retained successor result or verification file exists in the repository.

# 12 Discussion

## 12.1 What is invariant and what is representation-dependent

The retained projector and complete pointwise spectrum are invariant under the declared right-unitary attack. The represented matrix field, its block coefficients, and the spectra of its finite-support truncations are not. This distinction is the main scientific result. It prevents a band-only comparison from being interpreted as a complete statement about representation efficiency.

The pointwise oracle is especially important. Its recovery of the reference channel at roundoff confirms that the longer attacked tail is not caused by changing the retained subspace or losing spectral information. The global residual shows a different fact: one restricted constant action cannot reproduce a momentum-dependent inverse. These two alignment families must not be merged into one generic "alignment error."

## 12.2 Why complete-transform closure precedes locality claims

All three complete transforms reconstruct their construction matrix fields at roundoff. The range study therefore compares deliberate finite-support projections of closed finite transforms. Without this benchmark, a large attacked range error could be caused by transform normalization, representative ordering, cyclic Hermiticity, or incomplete reconstruction. The ordered protocol excludes those alternatives before interpreting locality.

The disjoint evaluation mesh provides an additional safeguard. Construction errors measure the finite transform on the data used to build it; evaluation errors probe newly sampled fields. Direct

evaluation targets separate complete-interpolant discrepancy from deliberate support deletion and their total consequence. The removed cyclic Nyquist block remains a fourth, power-valued convention channel rather than an off-node spectral error. Interpolation remains at roundoff while the attacked truncation channel follows the displayed range trend. This supports the finite truncation interpretation; it does not prove a continuum error bound.

### 12.3 Relation to the scalar control

The rank-one controlled-four-state report establishes that spectral reduction alone can produce a scalar Fourier tail even though scalar $U(1)$ conjugation is exactly inert. The present rank-two control adds the missing internal action: a nonconstant $U(2)$ right rotation redistributes the represented matrix field by the block convolution in Equation (29). The scalar and rank-two reports therefore form a mechanism-separation pair:

- rank one isolates spectral reduction, interpolation, and truncation without gauge-dependent represented-operator locality;
- rank two holds the retained subspace and complete spectrum fixed while activating momentum-dependent internal frame structure.

This comparison is methodological support for the scientific lineage of Ragasa2026a. The repository's current allocation ledger assigns the direct composite-alignment family to Ragasa2026c rather than the active Ragasa2026a evidence graph. Changing that publication allocation or binding these bytes into a manuscript manifest is a separate reviewed action; the technical report does not silently perform it.

### 12.4 Historical M2 and current canonical conventions

The named M2 parity values show continuity with the historical finite control while the range-four evaluation difference exposes a real convention boundary. Treating the historical and direct bytes as identical would require a compatibility transformation that obscures the current owners. The report instead preserves both values and explains why the direct source uses its own canonical transport and centered transform.

## 13 Limitations and nonclaims

The finite control does not establish:

- a continuum limit or a globally certified retained/excluded gap;
- uncertainty quantification or a forward-error bound for every reported digit;
- robustness under changes of parent coefficients, construction size, evaluation inventory, attack family, support rule, or tolerance;
- a universal exponential-localization law or universal hopping-range inflation;
- material applicability, density-functional-theory accuracy, or successful Wannierization;
- a topological interpretation of the closure eigenphases;
- higher-rank real or complex gauge behavior;
- an application threshold or a selected scientifically adequate range;

- publication acceptance, retained evidence, release, sealing, or a Project Koios–Formal certificate.

The one-global-unitary residual is diagnostic of the declared restricted family only. It is not a distance between physical systems. The attacked and reference complete fields remain pointwise unitarily equivalent.

The M3 admissible-set consumers now admit digest-bound canonical composite-result bytes and explicit verification dependencies. After that migration passed review, the live M2 runtime was retired without a compatibility facade. Its final source bytes and historical calculation artifacts remain recoverable provenance; they are not successor runtime inputs.

## 14 Conclusions

Within the 64-point construction and exactly disjoint 257-point evaluation control, the retained rank-two parent space is sampled-separated and the polar transport overlaps are well conditioned. The prescribed momentum-dependent right action preserves the retained projector and complete pointwise spectra while substantially changing the represented matrix field. Pointwise Procrustes alignment recovers the reference frame, projector, known inverse, and represented operator at roundoff. A single global unitary cannot recover the nonconstant attack.

The complete cyclic transforms close at roundoff, their negative-Nyquist conversion powers are separately negligible, and the complete ordinary interpolants agree with their direct targets at roundoff. The reported attacked range trend is therefore in the deliberate-truncation channel rather than transform failure, convention conversion, or finite interpolation. Reference and pointwise-aligned blocks decay rapidly and reach roundoff-scale range-eight errors. The attacked blocks have a much longer tail and retain an evaluation error near $7.04\times10^{-5}E_{\mathrm{ref}}$ at range eight. Exact pointwise band equivalence therefore does not determine finite block-hopping locality without a declared frame and support convention.

The full technical record contains the analytic mechanism, seven benchmarks, complete finite range inventory, independent reconstruction, historical parity boundary, literature position, provenance, and explicit nonclaims. That is the upstream level from which publication prose should be distilled. Retention or promotion still requires independent review, separately authorized direct execution, and the repository provenance and sealing gates.

Table 5: Declared controls for the provisional rank-two calculation.

| Control | Value |
|---|---|
| Parent / retained dimensions | 4/2 |
| Construction / evaluation sizes | 64/257 |
| Evaluation shift | 1/65 of one evaluation step |
| Ranges | $0, 1, 2, 3, 4, 6, 8$ |
| Attack angle | $0.30 + 0.65\sin(2\pi k) + 0.30\sin(4\pi k)$ |
| Gap threshold | $0.5E_{\mathrm{ref}}$ |
| Overlap singular-value threshold | 0.8 |
| Absolute / relative rank controls | $10^{-14}/10^{-12}$ |
| Energy / dimensionless tolerances | $10^{-11}E_{\mathrm{ref}}/10^{-12}$ |
| Cyclic / ordinary representatives | $-32, \ldots, 31$ / $-31, \ldots, 31$ |
| Error channels | conversion power; interpolation, truncation, total |
| Locality channels | reference, attacked, pointwise |
| Diagnostic-only alignment | one global unitary |

# A Frozen control inventory

# B Interpretation map

| Object | What it establishes | What it does not establish |
|---|---|---|
| Sampled gap | Finite retained/excluded separation | Global isolation certificate |
| Overlap singular value | Finite transport conditioning | Canonical backend-independent frame |
| Projector defect | Retained-space agreement | Represented-matrix agreement |
| Pointwise oracle | Known inverse recovery | Restricted global recovery |
| Global Procrustes residual | Best declared constant action | Physical inequivalence |
| Complete transform closure | Finite transform correctness | Locality or adequacy |
| Conversion power | Removed cyclic Nyquist weight | Off-node spectral error |
| Interpolation error | Complete interpolant versus direct field | Deliberate support loss |
| Truncation error | Finite range versus complete ordinary field | Total consequence |
| Total error | Finite range versus direct field | Sum of independent scalar errors |
| Omitted block norm | Ordinary coefficient-tail norm | Universal spectral-error law |
| SHA-256 digest | Exact byte identity | Scientific validity or sealing |

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